Solving the static Einstein equations with physics-informed neural networks

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A classical existence theory guarantees that the stationary vacuum Einstein equations, posed in the region outside a ball, have a unique solution for boundary data close enough to flat space, depending continuously on the data. We present a physics-informed neural network (PINN) solver for this boundary-value problem in the static case. The unknowns are a Riemannian 3-metric h and the squared norm λ of the static Killing vector, in harmonic gauge; the data are λ and the induced metric on an inner sphere, and asymptotic flatness is imposed by Robin conditions on an outer sphere. The connection is the Christoffel symbol of the network metric. We test the solver on Schwarzschild in harmonic coordinates and on two-rod Weyl solutions, on the symmetry axis and rotated so that no symmetry is left. For quadrupolar data, which have no closed-form solution, we compare with an independent spectral solution of the same problem. The harmonic gauge and the inner data leave the asymptotic chart free to translate, and this translation dominates the error until it is fixed by Cederbaum's pseudo-Newtonian center of mass, a conserved flux of the static equations; with it, the solution agrees with the spectral one to five to six orders below the size of the perturbation. Without symmetry the chart is also free to rotate; a second conserved flux, the antisymmetric counterpart of the first, fixes the rotation and lowers the error of the general three-dimensional network by three orders of magnitude, to the accuracy of the axisymmetric one. Finally, we read the Geroch–Hansen mass moments of the computed solutions off the symmetry axis, without constructing Weyl coordinates; they reproduce the exact moments of the Weyl solutions and the spectral-reference moments of the quadrupolar one.

I. INTRODUCTION

For many years there has been a mathematical theory of the stationary Einstein equations in an asymptotic region of spacetime, namely the region outside a ball [1]. It states that boundary data close enough to those of flat spacetime, together with the asymptotic conditions, produce a unique solution that depends continuously on the data. Very few attempts have been made to explore this theory numerically [2]. Here we show that this is possible, and that it opens a new window on these spacetimes.

The method also gives a simple way to explore other systems of possible physical importance: systems in which one solves for a Riemannian metric together with other fields, through equations that contain its Ricci tensor. Examples are the construction of minimal-energy data sets [3] and the equations that give higher-order Newtonian-limit expansions [4]. [TODO: more examples]

We restrict this work to the static Einstein equations. The stationary case is no harder from the point of view of the approximation method, and is left for a continuation of this work.

Physics-informed neural networks [5] represent the unknown fields by a neural network and minimize the squared residual of the equations and boundary conditions over sampled points. They need no mesh, handle the non-compact exterior through a logarithmic radial variable, and treat a configuration with no symmetry exactly as they treat a symmetric one. A small residual does not by itself mean a small error, so every result below is graded against an exact solution or an independent reference.

Section II states the equations, the gauge, the boundary and asymptotic conditions, and the residual gauge freedom. Section III describes the solver. Section IV tests it on exact solutions, Sec. V solves the quadrupolar problem, and Sec. VI extracts the multipole moments. Appendix A lists what is needed to repeat every computation.

II. THE STATIC EINSTEIN EQUATIONS

A. Variables and field equations

We write a static vacuum spacetime as

$$g = -\lambda\,\mathrm{d}t^2 + \lambda^{-1}h, \tag{1}$$

where h is a Riemannian metric on the spatial slice and $\lambda = -\xi\cdot\xi$ is the squared norm of the static Killing vector $\xi = \partial_t$; if $g = -N^2\mathrm{d}t^2 + \gamma$, then $\lambda = N^2$ and $h = N^2\gamma$. The vacuum equations become

$$R_{ab}(h) = \frac{1}{2\lambda^2}\,\nabla_a\lambda\,\nabla_b\lambda, \tag{2}$$

$$h^{ab}\nabla_a\nabla_b\lambda = \frac{1}{\lambda}\,h^{ab}\nabla_a\lambda\nabla_b\lambda. \tag{3}$$

With $U = \frac{1}{2}\ln\lambda$ these read $\mathrm{Ric}(h) = 2\,\mathrm{d}U\otimes\mathrm{d}U$ and $\Delta_h U = 0$.

In some cases we shall solve for the first system of equations, in some cases for the second equivalent system, depending on the problem. The difference being that U is closer to the Newtonian potential, and λ is the squared norm of the Killing vector. Thus, for instance the first multipole moments of U are closer to the Newtonian ones.

There are differences, of course, and in fact one finds the $\frac{1}{\rho^2}$ term in λ but not in U.

B. Gauge

We impose harmonic coordinates on h,

$$\Gamma^i \equiv h^{jk}\Gamma^i{}_{jk} = S^i, \tag{4}$$

with $S^i = 0$; equivalently $\Delta_h x^i = 0$ for the Cartesian-like coordinates $x^i = \rho\, n^i$. For the Weyl tests we instead use a source built from the candidate's own metric,

$$S^i = (h_{\rho\rho} - 1)\, h^{ij}\partial_j \ln \rho_n, \qquad \rho_n = \big|x - (x\cdot n)\, n\big|, \tag{5}$$

with n the symmetry axis, which the Weyl chart of Sec. IV C satisfies exactly.

C. The spherical solutions

The spherically symmetric solutions are our first test (Sec. IV B) and the background of the quadrupolar problem (Sec. V). In the harmonic chart the spherically symmetric solutions are

$$h_{ij} = \Big(1 - \frac{R_0^2}{\rho^2}\Big)\delta_{ij} + \frac{R_0^2}{\rho^2}\, n_i n_j, \qquad \lambda = k\,\frac{\rho - R_0}{\rho + R_0}, \tag{6}$$

i.e. $h = \mathrm{d}\rho^2 + (\rho^2 - R_0^2)\,\mathrm{d}\Omega^2$. With $k = 1$ this is Schwarzschild in harmonic coordinates: its mass is $M_0 = R_0$, with areal radius $r = \rho + R_0$ and $\lambda = 1 - 2M_0/r$. The harmonic condition still allows the radial maps $\rho \to c_1\rho + c_2 F_2(\rho)$, $F_2 = 2R_0 + \rho \ln \frac{\rho - R_0}{\rho + R_0}$, fixed by the round inner metric and $h \to \delta$.

D. Boundary and asymptotic conditions

The domain is the shell $\rho_{\rm in} \le \rho \le \rho_{\rm out}$.

a. Inner sphere. We prescribe λ and the induced metric,

$$\lambda|_{S_{in}} = \lambda_{in}, \qquad h_{induced}|_{S_{in}} = h_{2D} \tag{7}$$

where λ_{in} is any positive smooth function defined on the inner boundary and h_{2D} is any positive definite 2-metric defined on that same boundary. In particular we shall later use:

$$\lambda = \lambda_0 + S_1\, n\cdot e + S_2\, P_2(n\cdot e), \qquad PhP = \frac{r_{\rm in}^2}{\rho_{\rm in}^2}\, P, \tag{8}$$

with $P = I - n\otimes n$, $r_{\rm in}$ the areal radius of the sphere in h and e an arbitrary constant vector on harmonic coordinates. No condition is imposed on h_{rr} or h_{rA}. In harmonic gauge these are determined by the solution itself, and prescribing them would leave no solution of the boundary-value problem except when the prescribed values happen to coincide with the solution's own, which are not known in advance. Numerically this shows up as a loss that stalls instead of decreasing: a spectral approximation of the quadrupolar problem (Secs. III 0 e and V; 6 Legendre and 24 Chebyshev modes) computed with $h_{rA} = 0$ imposed stalled at a loss of 1.8×10^{-5}, with a non-decaying $l = 2$ part, and the same approximation reached 3×10^{-9} once the condition was removed. Since the spectral approximation is unrelated to the networks, this floor is a property of the boundary-value problem, not of the approximation method.

In the traditional approach to the problem one would look for boundary conditions that would imply the enforcement of the harmonic gauge condition once the equations are solved. In our approach the harmonic gauge condition is imposed directly in the interior: since the connection is the Christoffel symbol of the network metric, $\Gamma^i = S^i$ is a condition on the first derivatives of h, and it enters the loss as one more residual, together with the field equations (Sec. III); the boundary conditions then only have to select a unique solution among those of the equations and the gauge condition. This considerably simplifies their choice, since no conditions are needed whose role is to propagate the gauge.

b. Outer sphere. Asymptotic flatness is imposed by Robin conditions of order n,

$$B_n[f] \equiv \prod_{i=0}^{n-1} \big(\rho\,\partial_\rho + p_f + i\big)\,(f - f_\infty) = 0, \tag{9}$$

which annihilate the decaying powers $\rho^{-p_f}, \ldots, \rho^{-p_f-n+1}$. The exponent p_f is that of the asymptotic expansion of an asymptotically flat static solution, $\delta h \sim \rho^{-2}$, and the condition is imposed on h; its connection, $\Gamma \sim \rho^{-3}$, gets no separate condition. The condition is not imposed on λ, because its multipolar expansion, $\lambda - 1 = -2M/\rho + 2M^2/\rho^2 - \ldots$, violates the condition $\lambda - 1 \sim \rho^{-1}$ at the next order: the monopole alone already contains ρ^{-2}. Applied to the Schwarzschild solution (6) at $\rho = 20$, the local decay rates are 2.00 for δh and 3.005 for Γ, but 0.952 for $\lambda - 1$ instead of 1. For the scalar we therefore impose the condition on the Geroch–Hansen potential [6, 7]

$$\mathcal{P} = \frac{1 - \lambda^2}{4\lambda} = -\tfrac{1}{2}\sinh 2U, \tag{10}$$

whose monopole $M/\rho + M^3/\rho^3 + \ldots$ has no ρ^{-2} term, unlike $\lambda = 1 - 2M/\rho + 2M^2/\rho^2 - \ldots$. The Robin conditions are chosen so that $\mathcal{P}_\infty = 0$, i.e. $\lambda_\infty = 1$, and $h_\infty = \delta$; we use the order $n = 3$ for both h and $\mathcal{P}$ (order 2 in the Schwarzschild test of Sec. IV B). These pointwise conditions on the outer sphere are all that is imposed there.

E. The residual gauge freedom and the center of mass

The harmonic condition fixes the chart only up to harmonic diffeomorphisms $x \to x+\xi$, $\Delta_h \xi = 0$. One of them is left free by our boundary and asymptotic data: $\xi = 0$ on the inner sphere and $\xi \to k_T$, (k_T a constant vector) at infinity. It keeps the chart harmonic and the inner data unchanged, and translates the asymptotic chart by k_T. Its effect on the scalar, $\xi \cdot \nabla\lambda$, vanishes at $\rho_{\rm in}$ and decays like ρ^{-2} asymptotically, so the outer conditions barely penalize it: in the run of Sec. V without the condition, which drifted by $k_{T,z} = -1.8\times10^{-3}$, the dipole of $\mathcal{P}$ at $\rho = 100$ is -1.1×10^{-7}, i.e. $M_{\rm ADM}k_{T,z}/\rho^2$, against 4×10^{-11} in the run with it.

To avoid this freedom (which does not change the geometry but does change the pointwise values of our fields and so makes the error determination harder) we fix it by requiring the center of mass to vanish. Since U and the coordinates x^i are both h-harmonic, Green's identity makes

$$D^i(S) = \oint_S \left(x^i\,\partial_\nu U - U\,\partial_\nu x^i\right) \mathrm{d}A_h \qquad (11)$$

the same on every closed surface S enclosing the inner sphere, and $D = 4\pi M k_T$ for a mass M displaced by k_T. This is the pseudo-Newtonian center of mass that Cederbaum introduced for static ("geometrostatic") isolated systems [8], written in our variables. It is quasi-local, obeys an addition theorem for several separated sources, coincides with the ADM and constant-mean-curvature centers of mass in suitable asymptotic coordinates, and converges to the Newtonian center of mass in the Newtonian limit [8, 9]. [1] We impose $F_T \equiv D/4\pi = 0$ (so $F_T = Mk_T$) on a set of spheres. This is a gauge choice, available for any data; Sec. V shows that without it the translation mode is the largest error of the computed solution.

a. Rotations. The translations are not the only harmonic diffeomorphisms left free: harmonic coordinates of an asymptotically flat manifold are unique only up to rigid motions at infinity [22], and the rotations are free as well. For a constant vector k_R the field $\xi^i = \epsilon_{ijk}k_R^j x^k\,(\rho_{\rm in}^3/\rho^3 - 1)$ (flat-space form, $\rho_{\rm in} = 1$ below; ϵ_{ijk} is the permutation symbol, so $\epsilon_{ijk}k_R^j x^k = (k_R \times x)^i$) is harmonic, vanishes on the inner sphere and tends to the rigid rotation $-k_R \times x$ at infinity. It leaves λ and the induced metric of the inner sphere unchanged, since ξ and its tangential derivatives vanish there, and changes h only by $\delta h_{ij} \simeq -3\,(n_i\epsilon_{jkl} + n_j\epsilon_{ikl})\,k_R^k n^l/\rho^3$, in the components h_{rA}, which are not data; asymptotically it decays like ρ^{-3}, so the outer conditions do not see it either. For data symmetric about an axis, an axisymmetric network excludes the mode; a general three-dimensional network does not. To fix it we use the antisymmetric counterpart of Eq. (11). The coordinates are h-harmonic functions, $\Delta_h x^i = -\Gamma^i = 0$, so Green's identity applied to a pair of them, as it was applied to x^i and U in Eq. (11), makes

$$F_R^k(S) = \frac{1}{16\pi}\,\epsilon_{kij} \oint_S \left(x^i\,\partial_\nu x^j - x^j\,\partial_\nu x^i\right) \mathrm{d}A_h \qquad (12)$$

the same on every closed surface enclosing the inner sphere. Here ϵ_{kij} is the constant permutation symbol: it does not refer to any metric, only labels the three antisymmetric pairs (ij) of coordinate functions, and each component of F_R is an identity between two scalars, like D; the geometry enters through ∂_ν and $\mathrm{d}A_h$ alone. In terms of the coordinate normal n and area element $\mathrm{d}A$ of the sphere, $\partial_\nu x^j\,\mathrm{d}A_h = \sqrt{h}\,h^{jl}n_l\,\mathrm{d}A$, which is how we evaluate both fluxes; the $\sqrt{h}$ comes from that change of area element, not from a volume form. F_R vanishes for the spherical solutions, and for the mode above $F_R = k_R$ to first order. We impose $F_R = 0$ on the same spheres as F_T, with the same weight. Like F_T, it is a gauge choice available for any data; we have not found it used for this purpose in the literature. With a gauge source, $\Gamma^i = S^i$ (Sec. IV C), the coordinates are no longer harmonic and the fluxes change between two spheres by the integral over the shell between them, $-\frac{1}{8\pi}\int \epsilon_{kij}x^i S^j \sqrt{h}\,\mathrm{d}^3x$ for F_R and $\frac{1}{4\pi}\int U\,S^i\sqrt{h}\,\mathrm{d}^3x$ for F_T; in the Weyl tests we impose on each sphere the fluxes of the exact solution (those of F_R vanish, by axial symmetry of the source).

III. THE SOLVER

a. Networks. A neural network is, for our purposes, a family of functions depending on parameters θ. We use multilayer perceptrons: from an input vector z_0, built from the point x, the network computes

$$\begin{aligned} z_k &= \tanh\bigl(W_k z_{k-1} + b_k\bigr), \qquad k = 1,\dots,D, \\ y &= W_{D+1}\,z_D + b_{D+1}, \end{aligned} \qquad (13)$$

with tanh acting on each component. The depth D is the number of hidden layers z_k, each with N_w components (the width), and θ collects the entries of the matrices W_k and the vectors b_k; a layer from $n_{\rm in}$ to $n_{\rm out}$ components has $(n_{\rm in}+1)\,n_{\rm out}$ of them. The inputs are $n = x/\rho$ and $t = \ln(\rho/\rho_{\rm in})/\ln(\rho_{\rm out}/\rho_{\rm in}) \in [0,1]$. The general network has the seven outputs $y = (h_{ij}, u)$, $i \le j$, with $\lambda = e^u$; the output layer is initialized small so that training starts near flat space. The derivatives of h and λ with respect to x are computed by automatic differentiation, i.e. by the chain rule through the layers of (13): they are exact up to rounding, with no finite differences. The connection is the Christoffel symbol of h computed in this way, so

[1] The asymptotic notions it is compared with go back to Regge–Teitelboim and Beig–Ó Murchadha [10, 11] and to the foliation-based definitions of Huisken–Yau, Metzger, Huang, Corvino–Wu and Eichmair–Metzger [12–17] (see Ref. [18]); Cederbaum's analysis builds on the asymptotics of static fields of Beig, Beig–Simon and Kennefick–Ó Murchadha [19–21].

the compatibility equation holds identically. A first-order variant outputs Γ as well and adds compatibility as a residual; on the same Schwarzschild problem it is three to four orders of magnitude less accurate, Sec. IV B. For axisymmetric problems there is a simpler system, namely we look for (h, λ) of the form,

$$h_{ij} = (1+a)\,\delta_{ij} + b\,n_i n_j + \tfrac{c}{2}\,(n_i z_j + z_i n_j) + d\,z_i z_j, \qquad \lambda = e^u, \tag{14}$$

with a, b, c, d, u functions of $(t, \mu = n \cdot z)$, and optionally of $\rho_{\rm in}/\rho$ (the decay feature). At width 20 and depth 6 the five layers between hidden layers have $5 \times 21 \times 20 = 2100$ parameters in every network, and the counts differ only in the first and last layers: the axisymmetric network, with the three inputs $(t, \mu, \rho_{\rm in}/\rho)$ and the five outputs (a, b, c, d, u), has $4 \times 20 + 2100 + 21 \times 5 = 2285$ (2265 without the decay feature), and the general one, with the four inputs (n, t) and seven outputs, $5 \times 20 + 2100 + 21 \times 7 = 2347$. We also use networks of width 40 and depth 6 (the rotated Weyl test, the off-axis problem and the largest amplitudes of Sec. V): the hidden layers then have $5 \times 41 \times 40 = 8200$ parameters, and the totals are $4 \times 40 + 8200 + 41 \times 5 = 8565$ for the axisymmetric network and $5 \times 40 + 8200 + 41 \times 7 = 8687$ for the general one. Spherically symmetric tests use $h = \alpha(\rho)\delta + \beta(\rho) nn$.

b. Loss. The parameters are found by minimizing a loss: a non-negative function of θ that vanishes exactly when the network fields satisfy the equations, the gauge and the boundary conditions at the sampled points. Its terms are the residuals, the left-hand side minus the right-hand side of each condition evaluated on the network fields: R_g for the three groups g of interior equations, the Ricci equation (2), the gauge condition (4) and the scalar equation (3); $r_{\rm in}$ for the inner conditions (8); $r_{\rm out}$ for the outer conditions (9); and the fluxes F_T (center of mass) and F_R (rotation) of Sec. II E on a set of spheres S. The loss is

$$L = \sum_g w_g\, \overline{(\rho^{p_g} R_g)^2} + w_{\rm in}\overline{r_{\rm in}^2} + w_{\rm out}\overline{r_{\rm out}^2} + w_{\rm com} \sum_S |F_T|^2 + w_{\rm rot} \sum_S |F_R|^2, \tag{15}$$

where the bar is the mean over the sampled points, $\overline{F} = N^{-1}\sum_{k=1}^{N} F(x_k)$, and a square is the sum of the squares of the components. $p = (2, 1, 2)$ makes every residual dimensionless; for the Weyl and quadrupolar problems, in order to have a more accurate far field, we use $p = (4, 3, 4)$, an extra weight ρ^4 that determines the far field. $w_{\rm in} = w_{\rm out} = w_{\rm com} = w_{\rm rot} = 100$; the rotation term (12) is used only with the general three-dimensional network, in the runs of Secs. IV D and V C where it is said. The center-of-mass term uses a fixed product quadrature on each sphere, Gauss–Legendre in $\mu = \cos\theta$ and uniform in ϕ: 24×8 nodes for the axisymmetric network and 48×32 for the three-dimensional one (with data about an axis other than z, the coarser rule aliases the angular structure on the inner sphere; 48×32 agrees with 96×64 there).

c. Sampling points Interior points are drawn with uniform directions and logarithmically uniform radius ρ, i.e., with a density proportional to ρ^{-3}, which concentrates them near the inner boundary, where we expect the largest features of the fields. Boundary points are drawn uniformly on the inner and outer spheres. All points are redrawn every 500 iterations. Every result is checked on five fresh samples, on which the loss agrees with the training loss to within a few tens of per cent (the ratio is 0.95–1.18 for the runs we report; the two exceptions, the Weyl shell closest to the rods and the largest amplitude of the quadrupolar data, are noted where they appear). This checks that the parameters have not adapted to the particular points (overfitting): the network satisfies the equations between the points as well as at them.

d. Training. We minimize with two algorithms in succession: first the stochastic gradient method Adam [23] for 5000 steps, then the self-scaled Broyden quasi-Newton method SSBroyden [24, 25] with a dense inverse Hessian carried between blocks of 250 iterations, a strong-Wolfe line search and scale-invariant curvature guards, in stages of 10000 iterations; a block whose result is not finite is rejected. For the quadrupolar problem with the axisymmetric network, Adam (3 minutes) and two such stages (11 minutes each) reach a loss of 1.3×10^{-11}. Each stage restarted the inverse Hessian. The width-60 runs of Sec. V C and the continuation of Sec. IV D use instead a single run with stopping rules: Adam (at most 5000 steps) stops when its best loss has fallen by less than 30% over the last 1000 steps, and SSBroyden (at most 40000 iterations, one inverse Hessian throughout) stops when its best loss has fallen by less than 10% over the last 4000 iterations, after which the best field seen is kept. Everything runs in double precision with JAX [26] on NVIDIA A30 GPUs; one value and gradient at 16384 points costs 0.048 s.

e. Spectral approximations. A problem without an exact solution needs an independent approximation against which to measure the error of the networks. Since the loss (15) depends only on the fields, it can be minimized over any other family of functions. For axisymmetric data even under $z \to -z$ we write the fields as the spherical solution (6) plus

$$\delta\lambda = \sum_l f_0^l P_l,$$
$$\delta h_{ij} = \sum_l \big[f_1^l P_l\,\delta_{ij} + f_2^l P_l\, n_i n_j + f_3^l P_l'\,(e_i n_j + n_i e_j) + f_4^l P_l''\,\big(e_i e_j - \tfrac{1}{2}(1-\mu^2) P_{ij}\big)\big], \tag{16}$$

with $P_l(\mu)$ the Legendre polynomials, $e = \hat{z} - \mu n$ and $l = 0, 2, \ldots, L$ (for $l = 0$ only f_0^0, f_1^0, f_2^0), and each $f_a^l(\rho)$ a sum of Chebyshev polynomials of $\ln\rho$, $f_a^l = \sum_{j<J} c_{aj}^l\, T_j(2t-1)$, with $t = \ln(\rho/\rho_{\rm in})/\ln(\rho_{\rm out}/\rho_{\rm in}) \in [0, 1]$ as for the networks. The coefficients c minimize the same residual vector as the networks, on a fixed set of

points, by the Gauss–Newton method: with $r(c)$ the vector of all residuals and $J = \partial r/\partial c$, $c \to c - (J^T J)^{-1} J^T r$, which converges in a few iterations for so few unknowns. Apart from the residual that it minimizes, this method has nothing in common with the networks, so an agreement between the two is not an artifact of either. It also shows how small the loss can become for a given problem, and whether a stall belongs to the problem or to the representation: a stall of the spectral approximation, which has many parameters and a different optimizer, points to the problem itself, as when h_{rA} was prescribed on the inner sphere (Sec. II D). For data about another axis, the same approximation rotated onto that axis is the spectral reference against which the off-axis runs are graded. The size of the basis used for the quadrupolar problem is given in Sec. V.

IV. TESTS WITH EXACT SOLUTIONS

A. The pipeline on exact solutions

Before training anything we check the implementation itself. The loss (15) is computed from the values of h and λ and of their first and second derivatives at the points, and the code that computes it accepts any function $x \mapsto (h_{ij}(x), \lambda(x))$, not only a network (13). We give it the closed-form fields of an exact solution instead, for the spherical family (6)

$$x \mapsto \Big(\big(1 - \tfrac{R_0^2}{\rho^2}\big)\delta_{ij} + \tfrac{R_0^2}{\rho^2}\, n_i n_j,\ \tfrac{\rho - R_0}{\rho + R_0} \Big), \qquad (17)$$

and evaluate the loss through the same code: the derivatives by the same automatic differentiation, and the same residuals, boundary terms and weights. No network is trained or fitted in this check, and the exact solution is not converted into a network: the closed-form functions take the place of the network, and automatic differentiation differentiates them directly, as it would a network. The loss does not measure a difference from the exact solution; it measures how well the fields satisfy the equations, the gauge and the boundary conditions, so for an exact solution every term must vanish to rounding. A mistake in a sign or an index anywhere in the equations, the gauge or the boundary operators would leave a residual of order one. For (6) the total loss is below 10^{-16}. For the two-rod Weyl solution it is 1.3×10^{-28} with the gauge source (5); with the harmonic condition ($S = 0$) it is not small, since the Weyl chart is not harmonic, which shows that the check is sensitive to the gauge term. The rotated Weyl solution solves the system to 3×10^{-14} or better at every residual.

B. Schwarzschild in harmonic coordinates

We check the algorithm with the Schwarzschild solution of unit mass ($M_0 = R_0 = 1$). For that we use the spherical network with $R_0 = 1$, $\lambda_0 = 0.381966$ on the sphere of h-areal radius 2 and Robin conditions of order 2 on h and $\mathcal{P}$ at $\rho_{\text{out}} = 100$; the first-order network, which has an equation for Γ, gets the corresponding order-2 Robin condition on Γ as well. We recover the exact solution. The Robin condition is not set to zero, but rather to the value the exact solutions would produce, so it has a source term. Table I compares the metric-only and first-order formulations on the problem. The metric-only network is more accurate by three to four orders of magnitude (a factor 11000 in h, 1300 in Γ and 700 in λ). The first-order network's error is a gauge mode: along it the compatibility residual is nearly flat, and an earlier, less converged first-order run found the same geometry in a different harmonic chart, with h_{rr} off by a third. Deriving Γ from h removes the mode. With the homogeneous Robin conditions (no source), the metric-only network reaches a loss of 1.1×10^{-11} and maximum errors of 9.5×10^{-7} in h, 3.8×10^{-6} in Γ and 2.1×10^{-6} in λ against the exact solution, and its ADM mass $M_{\text{ADM}} = 1.00002$ (Sec. VI) against $M_0 = R_0 = 1$; each run takes 7–9 minutes on one GPU. The residuals here carry the weights $p = (2, 1, 2)$, without the extra ρ^4. With these weights, order 3 makes the outer term stiffer and leaves errors of 2×10^{-4} at the same loss; order 3 with $p = (4, 3, 4)$ gives 2.9×10^{-6} in h, still above order 2. (The center-of-mass condition plays no role here: a spherically symmetric network has no dipole, and the first-order network's error is a radial chart mode.)

TABLE I. First-order against metric-only network on the same Schwarzschild problem (manufactured Robin source of order 2, $\rho_{\text{out}} = 100$, 20000 Adam + 2000 SSBroyden iterations).

	first order (h, Γ, λ)	metric only (h, λ)
final loss	6.9×10^{-10}	6.9×10^{-12}
$\max \lvert\delta h\rvert$	1.0×10^{-2}	8.9×10^{-7}
$\max \lvert\delta \Gamma\rvert$	5.2×10^{-3}	4.1×10^{-6}
$\max \lvert\delta \lambda\rvert$	7.7×10^{-4}	1.1×10^{-6}

C. Weyl two-rod solutions

The Weyl class gives exact static axisymmetric solutions with nontrivial multipoles. Static axisymmetric vacuum metrics can be written in Weyl's canonical form [27],

$$g = -e^{2U}\mathrm{d}t^2 + e^{-2U}\left[e^{2k}(\mathrm{d}\rho_c^2 + \mathrm{d}z^2) + \rho_c^2 \mathrm{d}\phi^2\right], \qquad (18)$$

where ρ_c is the cylindrical radius, U is an axisymmetric solution of the flat Laplace equation, and k follows from U by a line integral, $\partial_{\rho_c} k = \rho_c\,[(\partial_{\rho_c} U)^2 - (\partial_z U)^2]$, $\partial_z k = 2\rho_c\, \partial_{\rho_c} U\, \partial_z U$, with $k \to 0$ at infinity. In our variables $\lambda = e^{2U}$ and $h = e^{2k}(\mathrm{d}\rho_c^2 + \mathrm{d}z^2) + \rho_c^2 \mathrm{d}\phi^2$. A rod of length L on the axis, $U_{\text{rod}} = \frac{1}{2}\ln[(R_a + R_b - L)/(R_a + R_b + L)]$

TABLE II. Weyl rods (masses 1 and 0.1), inner sphere moved towards the rods; Adam 5000 steps, then 2×10000 SSBroyden iterations; $d_{\rm hor}$ is the distance from the inner sphere to the nearest horizon; lengths in the units in which the rods have masses 1 and 0.1.

shell	$d_{\rm hor}$	$R_{ab}R^{ab}(\rho_{\rm in})$	$\max\lvert\delta h\rvert$	$\max\lvert\delta\lambda\rvert$	loss
$[7.5, 75]$	5.0	3.3×10^{-6}	2.7×10^{-7}	4.3×10^{-9}	1.1×10^{-14}
$[4, 40]$	1.5	5.8×10^{-3}	2.9×10^{-7}	7.3×10^{-9}	1.7×10^{-14}
$[3.5, 35]$	1.0	5.3×10^{-2}	4.7×10^{-7}	7.7×10^{-9}	7.2×10^{-14}

with R_a, R_b the distances to its ends, is the Schwarzschild solution of mass $L/2$; two rods give two black holes on the axis (held apart by a strut, where $k \neq 0$ between them). We evaluate k by Gauss–Legendre quadrature. The Weyl chart is not harmonic, but it satisfies the gauge condition (4) with the source (5), which is what we impose in these tests.

The inner boundary is the coordinate sphere $\rho_{\rm in} = 7.5$, three times the distance from the origin to the ends of the rods, and the outer one is at $\rho_{\rm out} = 75$. The inner data, λ and the induced metric, are those of the exact solution, and so are the sources of the Robin conditions; nothing else is taken from the exact solution. The runs use the formulation and the training of the quadrupolar problem of Sec. V: the axisymmetric network of width 20, 16384 points, Adam and two SSBroyden stages, in a chart scaled so that $\rho_{\rm in} = 1$. The fields are invariant under that scaling, and we quote lengths and moments in the original units. The center-of-mass condition is imposed with the value of the exact solution, $F_T = D/4\pi$ evaluated on it, instead of zero, since the Weyl chart of unequal rods is not mass centered. For two equal rods (masses 1, gap 1) on the shell $[7.5, 75]$ the network reaches a loss of 6.0×10^{-15}, with maximum errors 1.6×10^{-7} in h and 4.7×10^{-9} in λ (Fig. 1). Unequal rods (masses 1 and 0.1) give 1.1×10^{-14}, 2.7×10^{-7} and 4.3×10^{-9}. Their moments are in Sec. VI. To check the algorithm on a more non-linear region we move the inner sphere towards the rods: this raises the curvature at the inner sphere by four orders of magnitude, while the maximum error grows by less than a factor of two (Table II). On the closest shell one of the five fresh samples has a loss 50 times the training loss (the other four agree with it to 13%), a sign that the points do not resolve all of the strong-field region there.

D. A rotated Weyl configuration

The above approximations were obtained with a scheme adapted to the axis symmetry. In order to test the 3D setting we rotate the closest configuration by 45° about the y axis, so that the symmetry is not apparent, and use the generic 3D setting. The equations are covariant, so $h \to R\,h\,R^T$ and $\lambda \to \lambda$, evaluated at $R^T x$, is again a solution, and the inner data are taken from the rotated fields. The gauge, however, is a statement about

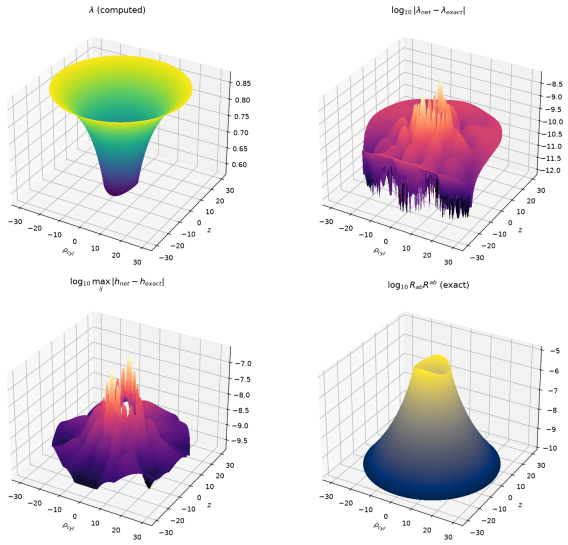


FIG. 1. Equal-mass Weyl rods on the shell $[7.5, 75]$, drawn over the meridional plane for $\rho \leq 30$ (the inner sphere is the inner edge; beyond $\rho = 30$ the fields are featureless): computed λ; $\log_{10}|\lambda_{\rm net} - \lambda_{\rm exact}|$ and $\log_{10}\max_{ij}|h_{\rm net} - h_{\rm exact}|$, floored four decades below their maxima so that the zero crossings of the error do not show as spikes; $\log_{10} R_{ab}R^{ab}$ of the exact solution.

a chart, and the source (5) must be taken about the rotated axis $n = R\hat{z}$: with that source the exact rotated solution satisfies the gauge condition to 2×10^{-15}, while it violates the harmonic one ($S = 0$) by $|\Gamma^i| \approx 2.6 \times 10^{-2}$. The general three-dimensional metric-only network, of width 40 with 32768 points, Adam and four SSBroyden stages (as for the off-axis data of Sec. V), reaches a loss of 1.3×10^{-11} (fresh/training ratio 1.18), with maximum errors 3.3×10^{-6} in h and 1.2×10^{-7} in λ, against 4.7×10^{-7} and 7.7×10^{-9} for the axisymmetric network on the same shell.

The general network is about seven times less accurate than the axisymmetric one in h and fifteen times in λ for the same problem, although it is four times larger and uses twice as many points: it has to represent functions of three variables instead of two. Both the size of the network and the number of points matter there (Sec. V): the loss is a mean over the sampled points, and with too few of them a larger network can lower the mean while the residual grows at points it does not sample.

Part of that difference is the rotation mode of Sec. II E: fitted on spheres, it accounts for 50–70% of the error in h for $\rho \leq 3$. The identities behind the two fluxes hold for this network to the order of its residuals: F_T agrees with the flux of the exact solution to 2×10^{-9}, and F_R, $(0.4\text{–}1.1) \times 10^{-6}$ instead of zero, changes between spheres by the volume integral of the gauge residual. Continuing the run with the rotation condition (Fig. 5, right), SSBroyden stops after 5250 iterations at the same loss, 1.4×10^{-11}, with the rms error in h lowered from 5.3×10^{-7} to 2.0×10^{-7}, while its maximum (3.1×10^{-6})

and the error in λ (1.5×10^{-8} rms) do not change.

V. QUADRUPOLAR DATA

Having tested the solver on exact solutions, we now apply it to a problem without one: spherical data with a quadrupolar term.

A. The problem

We take the inner sphere at $\rho_{\rm in}=1$, with h-areal radius $r_{\rm in}=1$ and $\lambda_0=1/3$. The datum is

$$\lambda|_{\rho_{\rm in}} = \tfrac{1}{3} - \tfrac{1}{12}\,P_2(\cos\theta), \tag{19}$$

a 25% perturbation of the spherical value, with no known closed-form solution. The constants λ_0, S_2 and $r_{\rm in}$ are free data, and the scale of λ is fixed by the outer condition $\lambda\to1$. For $S_2=0$ the data describe a Schwarzschild solution whose sphere has Schwarzschild radius $r_S=r_{\rm in}/\sqrt{\lambda_0}$ (the areal radius in $h=\lambda\gamma$ is $\sqrt{\lambda}\,r_S$) and $\lambda_0=1-2M_0/r_S$, so that $M_0=r_{\rm in}(1-\lambda_0)/(2\sqrt{\lambda_0})=R_0=1/\sqrt{3}$; this is the spherical (Schwarzschild) reference used below. [2]

We fix the outer boundary at $\rho_{\rm out}=100$ so that the fields there are already at the asymptotic regime. Linear theory predicts a pure $l=2$ response decaying like ρ^{-3}, no $l=1$ part and an $l=0$ back-reaction of order S_2^2. Against the spherical solution the perturbation has an rms size of 2.0×10^{-2} in λ and 2.2×10^{-2} in h near the inner sphere, and 2.6×10^{-4} and 3.5×10^{-6} for $30<\rho<100$.

We grade against a spectral approximation (16) (Sec. III 0 e) with $L=10$ and $J=36$, 1008 unknowns, computed on 2500 points with the same losses, weights and boundary conditions of the networks, in five Gauss–Newton steps (6 minutes). Its loss is 1.6×10^{-13}. Its $l=2$ part decays as $\rho^{-3.00}$ from $\rho=20$ to 100, as linear theory predicts. A larger basis, $L=14$ and $J=48$, changes it by 1.7×10^{-8} in h and 1.9×10^{-9} in λ (rms over the shell), ten times less than the errors of the best networks below, and grading against it changes those errors by less than 20%. The errors we quote are therefore errors of the networks, measured by an approximation obtained by an unrelated method. Its mass, and how the mass depends on the datum, are in Sec. VI.

a. The geometry of the network solution. To see how strongly the datum $S_2=-1/4$ (λ from 0.083 at the poles of the sphere to 0.458 at the equator) deforms the geometry, we evaluate the gauge-invariant curvature of the width-20 network in the cylindrical chart about the axis, for the static vacuum spacetime $g=-\lambda\,\mathrm{d}t^2+h/\lambda$ (Fig. 2). The Kretschmann scalar K is of order one at the sphere (7.9 on the axis and 2.2 at the equator, for $\rho=1$) and falls as ρ^{-6}, by twelve orders of magnitude between $\rho=1$ and $\rho=100$. Against the spherical (Schwarzschild) reference with the same λ_0 it is 13 times larger on the axis at the sphere (a maximum of 14.6 at $\rho\simeq1.3$), but at the equator it goes from 3.6 times larger at the sphere to 0.14 at $\rho\simeq1.8$, a change by a factor 25 within 0.8 in ρ; beyond $\rho\simeq10$ the ratio tends to $(M_{\rm ADM}/M_0)^2=1.35$, the mass shift. The algebraic speciality $|S-1|$ ($S=1$ for type D, which the spherical reference is) is of order one at the equator of the sphere (1.0 at $\rho=1$, 0.29 at 3, 1.3×10^{-3} at 10), so near the sphere the solution is far from type D, and it decays as a power of ρ. The network's four-dimensional Ricci tensor, which vanishes for an exact solution since we are in vacuum, has the norm $|R^{(4)}_{ab}|/\sqrt{K}=3.5\times10^{-5}$ (median) for $1\le\rho<3$, 6×10^{-6} for $3\le\rho<10$ and 7×10^{-7} beyond, and its curvature differs from that of the spectral reference by 7.6×10^{-5}, 5.0×10^{-6} and 1.3×10^{-6} in the same ranges (the largest value, 5×10^{-2}, is on the axis at the sphere).

The Hawking mass of the coordinate spheres (Fig. 3) rises from 0.500 at the sphere to 0.668 at $\rho=3.5$ and stays there, and the lapse mass $r_a(1-\lambda)/2$ approaches the same value from above (0.687 at $\rho=1$). Both are within 1.1×10^{-4} of the mass $M_{\rm ADM}=0.66961$ of the multipole extraction for $\rho\gtrsim8$, well inside the 1.1×10^{-3} spread of that extraction, while the spherical reference has the constant 0.577. The Hawking mass of the network differs from that of the spectral reference by 6×10^{-6} (relative to $M_{\rm ADM}$) at the sphere and by 3×10^{-7} beyond $\rho\simeq5$.

B. Without and with the center-of-mass condition

Two runs with identical settings, the formulation of Sec. II D, except for the center-of-mass term (Appendix A), show what the condition does (Table III, Fig. 4). Without it, the error is dominated by a term in the approximation that is odd under $z\to-z$ (an $l=1$ coefficient of λ peaking at 1.4×10^{-4} near $\rho\approx2$), which the even data forbid; every term of the loss is exactly reflection symmetric, so nothing forces this part: it is the translation mode of Sec. II E. The flux (11) confirms it: it is the same on the spheres $\rho=1,2,10$ and 100 but not zero, $F_{T,z}=-1.063\times10^{-3}$, and the λ error near the inner sphere is $0.061\,|F_{T,z}|$. With $F_T=0$ imposed on the spheres $\rho=1,1.5,3,10,30,100$, the error falls by a factor 5000 in λ and 2800 in h, to 7.1×10^{-9} and 1.45×10^{-7} (rms over the shell), six and five orders below the size of the perturbation. The loss does not see the difference: it is 1.6×10^{-11} without the condition and 1.3×10^{-11} with it, because the translation costs almost nothing in it. For the same reason its size without the condition is not fixed: it depends on how far the optimization happens to drift along the mode (the same run with extra conditions on the outer spheres drifted to $F_{T,z}=7.6\times10^{-5}$, 14

[2] That $\rho_{\rm in}$ and $r_{\rm in}$ are equal is not a contradiction: the harmonic gauge leaves the radial map $\rho\to\rho+c_2F_2(\rho)$ free (Sec. II), and the inner datum fixes c_2.

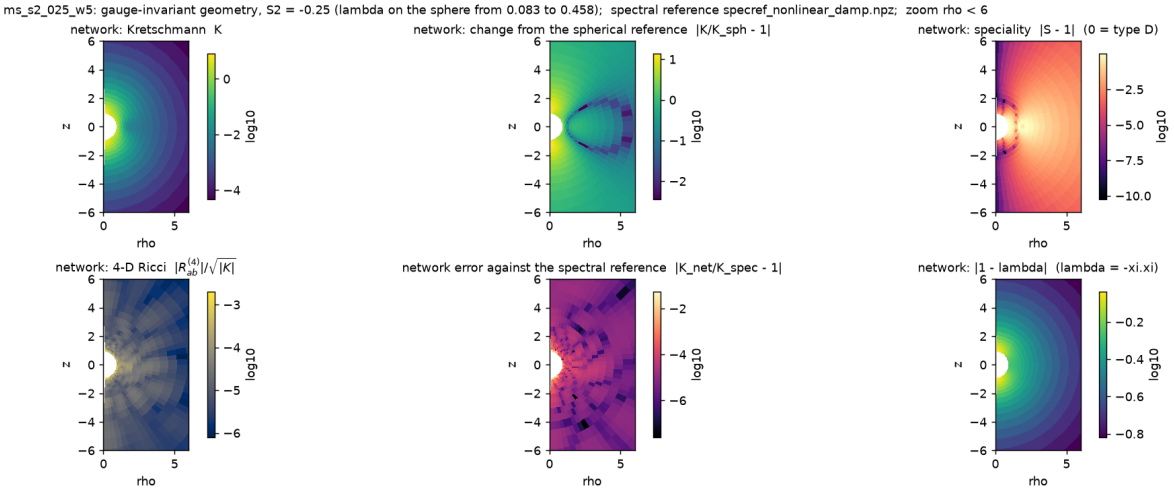


FIG. 2. Gauge-invariant geometry of the network at $S_2 = -1/4$, meridional plane, $\rho < 6$ (the sphere $\rho = 1$ is the white half-disk): its Kretschmann scalar K; its change from the spherical (Schwarzschild) reference, $|K/K_{\rm sph} - 1|$; its algebraic speciality $|S-1|$ (zero for type D); its four-dimensional Ricci tensor $|R^{(4)}_{ab}|/\sqrt{|K|}$; its error in K against the spectral reference; and $|1-\lambda|$. Logarithmic color scales.

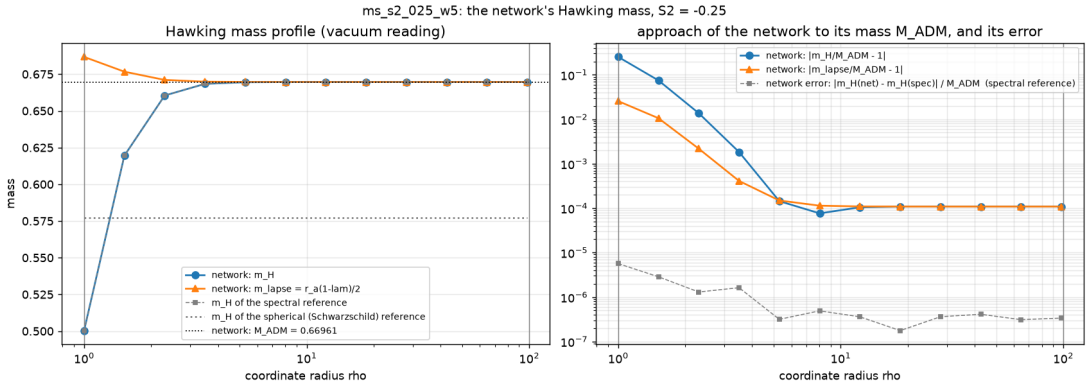


FIG. 3. Hawking mass and lapse mass of the network at $S_2 = -1/4$ on coordinate spheres, with its mass $M_{\rm ADM}$; the gray curves are the Hawking mass of the spectral reference and of the spherical (Schwarzschild) reference (left), and the deviations of the network from $M_{\rm ADM}$ and from the spectral reference (right).

times less). Two more runs of each kind, from different random initializations and samples, show this directly. Without the condition they drift to $F_{T,z} = -6.6 \times 10^{-3}$ and $+1.7\times10^{-3}$, with rms errors 2.5×10^{-3} and 6.4×10^{-4} in h, while their losses, 6.4×10^{-12} and 1.7×10^{-11}, are as low as before. With it the three runs agree: 1.45, 2.1 and 3.1×10^{-7} in h, and 7.1×10^{-9}, 1.7×10^{-8} and 1.9×10^{-8} in λ, with $|F_T| \leq 1.2 \times 10^{-7}$. The center of mass condition removes the mode.

C. Off-axis data

The same datum about the axis $(1, 0, 1)/\sqrt{2}$. Since we keep the same harmonic coordinates it looks like a fully three-dimensional problem whose solution is the rotated axisymmetric one. Here we solve it with the whole 3-dimensional network. We solve it with the general metric-only network in the formulation of Sec. II D, Adam and four SSBroyden stages. At network-width 20 (2347 parameters, 16384 points) the loss levels off at 1.2×10^{-6}, five orders above the axisymmetric one, decreasing by only 10–25% per stage, with rms errors of 9.1×10^{-6} in λ and 1.2×10^{-3} in h. Doubling either the width or the number of points alone gains little: width 40 at 16384 points lowers the loss to 1.4×10^{-8} but leaves the errors at 1.4×10^{-5} and 7.2×10^{-4}, and width 20 at 32768 points gives 5.4×10^{-7}, 1.9×10^{-5} and 8.2×10^{-4}. With both, width 40 (8687 parameters) and 32768 points, the loss falls to 4.3×10^{-9} within two stages and stays there (fresh/training ratio 1.00), and the error against the rotated spectral reference is 3.4×10^{-6} in λ and 2.6×10^{-4} in h (rms over the shell), i.e. 2800 and 40 times below the perturbation; a second run from a different initialization gives 3.6×10^{-6} and 3.3×10^{-4}. A larger network needs more points to constrain it. The error concentrates in h

TABLE III. Error against the spectral reference (angular rms) by radial region, for the same run without and with the center-of-mass condition (Adam 5000 steps, then 2 × 10000 SSBroyden iterations; the loss is the final training loss).

	loss	λ error				h error			
		1–3	3–10	10–30	30–100	1–3	3–10	10–30	30–100
perturbation size		2.0×10^{-2}	2.5×10^{-3}	7.6×10^{-4}	2.6×10^{-4}	2.2×10^{-2}	4.5×10^{-4}	3.1×10^{-5}	3.5×10^{-6}
without center of mass	1.6×10^{-11}	6.5×10^{-5}	3.2×10^{-5}	4.9×10^{-6}	5.9×10^{-7}	8.2×10^{-4}	1.2×10^{-4}	1.2×10^{-5}	1.2×10^{-6}
with center of mass	1.3×10^{-11}	1.2×10^{-8}	6.5×10^{-9}	3.7×10^{-9}	2.7×10^{-9}	3.0×10^{-7}	3.3×10^{-8}	2.5×10^{-9}	1.9×10^{-9}

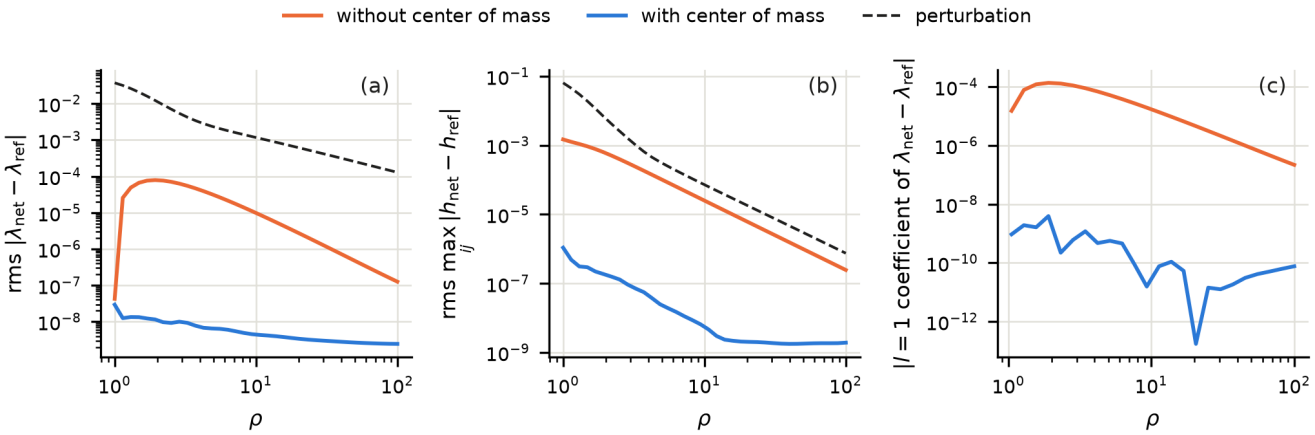


FIG. 4. Error against the spectral reference without and with the center-of-mass condition: angular rms of $\lambda_{\rm net} - \lambda_{\rm ref}$, of $\max_{ij} |h_{\rm net} - h_{\rm ref}|$, and the $l = 1$ coefficient of $\lambda_{\rm net} - \lambda_{\rm ref}$. Dashed: the perturbation (reference minus spherical solution).

near the inner sphere (5.3×10^{-4} for $1 \le \rho \le 3$, against a perturbation of 2.3×10^{-2}) and falls to 3×10^{-8} beyond $\rho = 30$; $|F_T| \le 7.5 \times 10^{-7}$ on every sphere. Each stage takes 30 minutes at width 40 (10 at width 20).

a. The rotation mode. Doubling both again, width 60 (19027 parameters) and 65536 points, in a single run (I), brings the loss down to 1.6×10^{-11}, as low as that of the axisymmetric network, but not the error: 8.5×10^{-6} in λ and 6.7×10^{-4} in h (Fig. 5). The loss and the error have come apart, and the reason is the rotation mode of Sec. II E. On the spheres $1.2 \le \rho \le 50$ one constant vector k_R, $|k_R| = 2.3 \times 10^{-3}$, accounts for 96–99% of the error in h, and the flux (12) of the network has the same size, $|F_R| = 2.3 \times 10^{-3}$, on every sphere from $\rho = 1$ to 100, while that of the spectral reference is below 2×10^{-12}. In the staged runs at width 40 the mode, $|k_R|$ between 1×10^{-3} and 2.5×10^{-3}, is set in the first SSBroyden stage and does not change while the loss falls by another order of magnitude: it costs almost nothing in the loss. Continuing I with the condition $F_R = 0$ (run L) removes it within 3000 iterations, $|k_R| < 10^{-9}$, and SSBroyden stops after 11000 iterations at the same loss, 1.6×10^{-11} (fresh/training ratio 1.03). The error falls to 7.6×10^{-9} in λ and 2.6×10^{-7} in h (rms; maxima 1.0×10^{-7} and 3.3×10^{-6}), 1100 and 2600 times below that of I and the same as the axisymmetric network on the aligned datum (Table III). In h it is 5.4×10^{-7} for $1 \le \rho \le 3$ and 1.5×10^{-9} beyond $\rho = 30$, and $|F_T| \le 1.4 \times 10^{-8}$. With both conditions, then, the general network solves the problem without symmetry as accurately as the symmetric network solves the symmetric one. Run L takes 2.2 hours on one A30.

b. The inner data in H^1. Since the error of every run concentrates at the inner sphere, a natural extension is to impose the inner data together with their derivatives along the sphere: we add to the loss $\rho\,(I - n \otimes n)\nabla$ of the λ residual and of the induced-metric residual $PhP - (r_{\rm in}^2/\rho^2)P$ of Eq. (8), with the weight of the inner term, so that the inner condition is imposed in the H^1 norm of the sphere rather than in L^2. On the sphere PhP is the induced metric h_{AB} alone, so the new terms involve only the data, and h_{rr}, h_{rA} remain free. On the axisymmetric problem, with the same budget, it lowers the loss from 1.3×10^{-11} to 9.4×10^{-12} and the error from 7.1×10^{-9} to 3.7×10^{-9} in λ and from 1.45×10^{-7} to 1.15×10^{-7} in h (by a factor 1.8 in λ for $1 \le \rho \le 3$). Over three initializations each, the error in λ is $(2.9\text{–}3.7) \times 10^{-9}$ with the term and $(0.7\text{–}1.9) \times 10^{-8}$ without it, and the one in h $(0.9\text{–}1.6) \times 10^{-7}$ against $(1.45\text{–}3.1) \times 10^{-7}$: the gain is larger than the scatter between runs. On the off-axis problem at width 40 it slows the training down: after the same four stages the field-equation residuals are seven times larger and the error is 1.6–1.8 times larger (6.1×10^{-6} in λ, 4.0×10^{-4} in h), with the loss still decreasing (2.9×10^{-8}) where the run without the term had leveled off. Two further stages bring it to 1.9×10^{-8}, now leveling off at 4.5

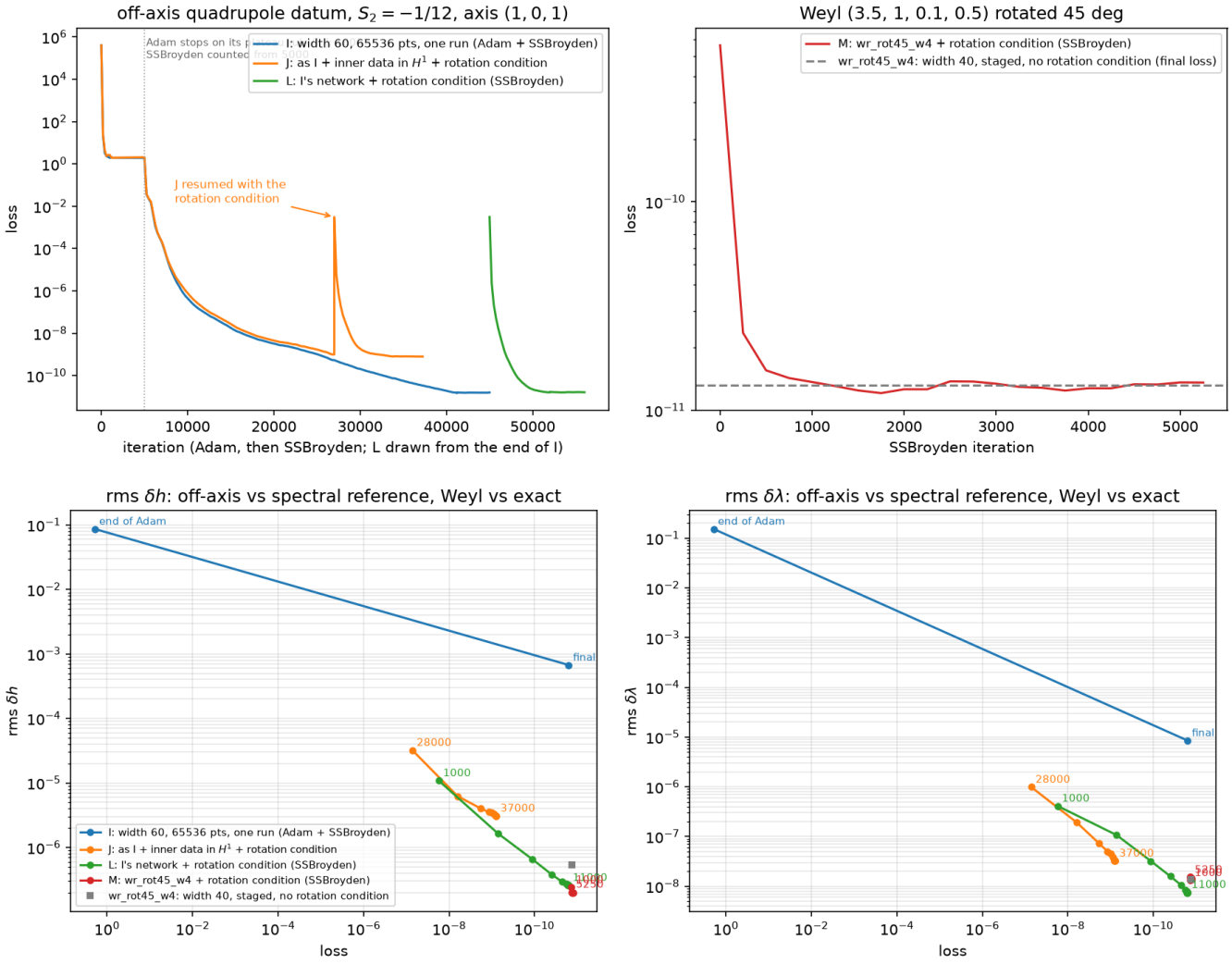


FIG. 5. Top: loss against iteration. Left, the off-axis datum with the width-60 network: I (no rotation condition), J (inner data in H^1; the rotation condition added at its iteration 27000, where the loss jumps), and L (the final network of I continued with the rotation condition, drawn from the end of I); Adam stops on its plateau rule near step 1600 and the SSBroyden iterations are counted from 5000. Right, the rotated Weyl network continued with the rotation condition, with the final loss of the run without it (dashed). Bottom: rms error against the loss along the same runs (snapshots every 1000 iterations; for I, the end of Adam and the final network), in h (left) and λ (right); off-axis runs against the rotated spectral reference, Weyl runs against the exact solution.

times the loss of the run without the term, with the error still 1.4–1.5 times larger (4.7×10^{-6} in λ, 3.8×10^{-4} in h). A larger network does not change this. At width 60 with 65536 points (run J, with the rotation condition added at its iteration 27000), the loss levels off at 7.9×10^{-10}, fifty times that of L, with field-equation residuals 25–70 times larger, and the error is 3.2×10^{-8} in λ and 3.1×10^{-6} in h, four and twelve times that of L (Fig. 5). In three dimensions the term therefore costs more than it gains, and we do not include it in the formulation.

VI. MULTIPOLE MOMENTS

a. Extraction from the axis. For a static axisymmetric vacuum solution the Geroch–Hansen moments [6, 7, 28] follow from the potential on the symmetry axis in canonical Weyl coordinates, $g = -e^{2U}\mathrm{d}t^2 + e^{-2U}[e^{2k}(\mathrm{d}W^2+\mathrm{d}Z^2)+W^2\mathrm{d}\phi^2]$: with the static Ernst potential $\xi = -\tanh U$ on the axis, $\xi(0,Z) = \sum_j m_j/|Z|^{j+1}$, the moments are $M_{\mathrm{ADM}} = m_0$ (the ADM mass), $M_j = m_j$ for $1 \le j \le 3$ and $M_4 = m_4 - \frac{1}{7}m_0(m_0m_2 - m_1^2)$ [29]; equivalently, from $U(0,Z) = \sum_r U^{(r)}/|Z|^{r+1}$, $M_{\mathrm{ADM}} = -U^{(0)}$, $M_1 = -U^{(1)}$, $M_2 = \frac{1}{3}(U^{(0)})^3 - U^{(2)}$, $M_3 = (U^{(0)})^2U^{(1)} - U^{(3)}$ [30]. Weyl co-

ordinates need not be built, because only the axis is used: in our variables $U = \frac{1}{2}\ln\lambda$ and $W = |\partial_\phi|_h$; on the part of the axis that reaches infinity $k = 0$, so there $\mathrm{d}Z$ is the proper h-length along the axis; the inner ball cuts the axis in two, and the offset between the branches is the change of Z along a meridian semicircle, $\mathrm{d}Z = \star\mathrm{d}W$ in the meridian metric. We fit ξ on both branches and fix the origin of Z by mass centring ($M_1 = 0$); the error bar is the spread over fit windows starting at 1–1.5 ρ_{in} and fit orders 6 and 8. On exact solutions this returns: Schwarzschild in harmonic coordinates, $M_{\mathrm{ADM}} = M_0 = R_0$ to 10^{-12} and M_l below 10^{-11} for $1 \le l \le 4$; the equal Weyl rods, $M_{\mathrm{ADM}} = 2$ and $M_2 = 2.500000$; the unequal rods, the closed-form moments to 10^{-5} relative and the center of mass $z = 1.309091$; Curzon, $M_2 = -M^3/3$ and $M_4 = 19M^5/105$, although U itself has no quadrupole.

b. The Weyl networks. The trained solutions give $M_{\mathrm{ADM}} = 2.000000$ and $M_2 = 2.49996$ for the equal rods (exact 2 and 2.5), and $M_{\mathrm{ADM}} = 1.100000$, $M_2 = 0.290910$, $M_3 = -0.49984$ with the center of mass at $z = 1.309091$ for the unequal rods (exact 1.1, 0.290909, −0.49983, 1.309091). For the rotated rods on the closest shell, extracted about the rotated axis, the three-dimensional network gives $M_{\mathrm{ADM}} = 1.100000$, $M_2 = 0.29093$, $M_3 = -0.4930$ and the center of mass at 1.30904. The axisymmetric network on the same shell gives the same values to five digits, so the remaining differences from the exact ones (1.4% in M_3) come from the extraction, whose fit windows start close to the rods on that shell, and not from the networks.

c. The quadrupolar solution. Table IV lists the moments. The data raise the mass from the value $M_0 = R_0 = 0.57735$ of the spherical reference to $M_{\mathrm{ADM}} = 0.583847$, a nonlinear back-reaction of 1.1% in the monopole, and the quadrupole moment is $M_2 = 0.1747$. The run without the center-of-mass condition, whose pointwise error is 5000 times larger, has moments as close to the spectral reference as the run with it (within 3×10^{-7} in M_2 and 3×10^{-6} in M_4, against an extraction spread of 1.2×10^{-4} and 7×10^{-4}): moments are geometric, and the translation mode moves the chart, not the geometry. The extraction shows the mode directly: the center of mass of that run, measured from the center of the coordinate sphere of radius ρ, is -8×10^{-7} at $\rho = 1$, -8.1×10^{-4} at $\rho = 2$, -1.63×10^{-3} at 10 and -1.79×10^{-3} at 70: it vanishes at the inner sphere, as the mode must, and tends to the translation $k_{T,z} = F_{T,z}/M_{\mathrm{ADM}} = -1.82 \times 10^{-3}$. With the condition it is below 6×10^{-8} at every radius.

d. The mass as a function of the datum. The mass shift $M_{\mathrm{ADM}} - M_0$ is the $l = 0$ back-reaction of the quadrupole, second order in S_2 according to linear theory. To check this we trained networks and solved the spectral reference for seven amplitudes (Table V, Fig. 6). For $|S_2| \le 1/6$ the mass shift is positive for both signs of S_2, and it depends little on the sign (at $S_2 = \pm1/12$ it is 6.50×10^{-3} and 5.57×10^{-3}). Doubling S_2 multiplies it by 4.2 (from $-1/24$ to $-1/12$) and by 4.7 (from $-1/12$ to $-1/6$), approaching the factor 4 of a quadratic law for small S_2. The fit $M_{\mathrm{ADM}} - M_0 = aS_2^2 + bS_2^3 + cS_2^4$ to the five spectral references with $a = 0.83$, $b = -0.77$, $c = 5.2$ reproduces them to 4×10^{-5}, and no term linear in S_2 is needed (the even part of the two values at $S_2 = \pm1/12$ alone gives $a = 0.87$). The quadrupole moment is linear to leading order, with M_2/S_2 going from -2.00 at $S_2 = -1/24$ to -2.43 at $-1/6$. The masses of the networks (width 20, Appendix A) agree with those of their spectral references to 2×10^{-9} at $-1/24$, 4×10^{-8} at $-1/12$ and 4×10^{-7} at $-1/6$, far below the shifts, and their errors scale with the data: loss 3.2×10^{-12} and rms errors 2.7×10^{-9} in λ, 1.0×10^{-7} in h at $-1/24$; 2.5×10^{-10}, 2.2×10^{-7} and 1.9×10^{-6} at $-1/6$, where λ ranges from 1/6 to 5/12 on the sphere.

Beyond that the shift stops being quadratic: $(M_{\mathrm{ADM}} - M_0)/S_2^2$ rises from 1.10 at $-1/6$ to 1.48 at $-1/4$ and 2.10 at -0.30, where λ falls to 1/12 and 1/30 at the poles of the sphere, and the fit of the small amplitudes underestimates the shift by 9% and 27% there. The hard end also asks more of both methods. At $S_2 = -1/4$ the spectral reference needs angular modes up to $l = 14$ and 48 radial modes (its residual is 4.6×10^{-5}), and the width-20 network then reaches a loss of 2.5×10^{-8} with rms errors 3.9×10^{-7} in λ and 7.0×10^{-6} in h and a mass 6×10^{-7} from that of the spectral reference. A width-40 network trained on 65536 points, redrawn every 250 iterations, lowers the loss to 2.2×10^{-10}, the same on fresh samples (ratio 0.95), and the errors to 2.0×10^{-7} and 3.3×10^{-6}, with a mass 6×10^{-7} and a quadrupole moment 2×10^{-6} from those of the spectral reference. With half as many points the same network reaches the same errors, but its loss on fresh samples is eleven times its training loss: a network of this size needs the larger sample. At $S_2 = -0.30$ the spectral reference needs $l \le 20$: with $l \le 14$ it stalls at a loss of 1.3×10^{-6}, and the corrected one (residual 7×10^{-5}) moves its mass by 5×10^{-4}. The networks do not converge there. The width-20 one stops at a loss of 1.6×10^{-6}, errors 4.7×10^{-5} and 8.1×10^{-5} and a mass 7×10^{-5} from that of the spectral reference. The width-40 one with 65536 points reaches 4.3×10^{-9} on its training points but 39 times that on fresh samples, with errors 1.3×10^{-5} and 1.6×10^{-5},

TABLE IV. Geroch–Hansen moments of the quadrupolar solution: the spectral reference, and the differences network − spectral reference with the same fit windows (the spectral reference's window spread, 1.2×10^{-4} in M_2 and 7×10^{-4} in M_4, largely cancels in the differences). For the rotated data the moments are taken about the data axis.

	M_{ADM}	M_2	M_4
spectral reference	0.583847	0.17471	0.022
with center of mass	-3.6×10^{-8}	$+2.4 \times 10^{-7}$	$+2.4 \times$
without center of mass	$+2.1 \times 10^{-9}$	-3.1×10^{-7}	$+2.8 \times$
3-D network (width 20), rotated data	-5.9×10^{-6}	$+2.4 \times 10^{-5}$	$-1.2 \times$
3-D network (width 40), rotated data	-4.0×10^{-7}	$+3.5 \times 10^{-6}$	$-3.6 \times$

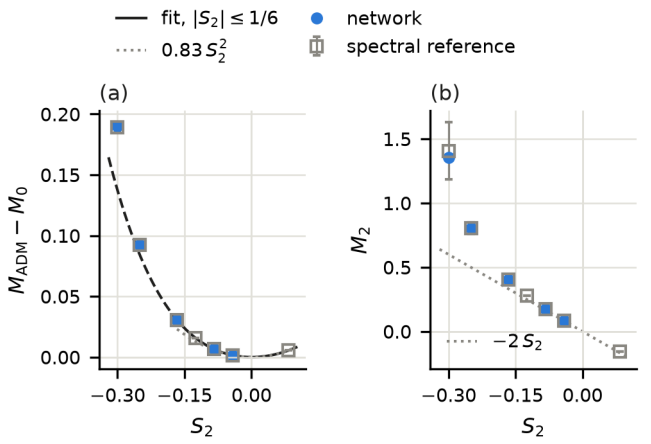


FIG. 6. The mass shift (a) and the quadrupole moment (b) against the amplitude S_2 of the datum (Table V): the masses of the networks (points) and of the spectral references (open squares, with the spread of the extraction). Solid: the fit $0.83S_2^2 - 0.77S_2^3 + 5.2S_2^4$ of the five spectral references with $|S_2| \le 1/6$, dashed beyond them; dotted: its quadratic term, and $-2S_2$.

a mass 9×10^{-4} away, within the 4×10^{-3} spread of the extraction of the mass, and M_2 4% off. With half as many points it had the same errors and a fresh/training ratio of 170: more points reduce the overfitting but not the error, so at this amplitude the limit is the network and its training, not the sample.

e. Decay of the multipoles. Expanding $\mathcal{P}$ in Legendre modes $c_l(\rho)$ about the solution's own axis, the local power $\mathrm{d}\ln|c_l|/\mathrm{d}\ln\rho$ of the spectral reference is -1.00, -3.00 and -5.0 for $l = 0, 2, 4$ (the last out to $\rho \approx 30$, where $c_4 \sim 10^{-10}$), and its odd modes are at 10^{-16}. The network with the center-of-mass condition gives the same powers, with its odd modes below 10^{-8}. The three-dimensional network on the rotated data, expanded about the data axis, has odd modes of 1.3×10^{-6} at width 20, with its $l = 2$ tail holding out to $\rho \approx 30$ and its $l = 4$ tail only to $\rho \approx 5$ (M_4 good to 5%); at width 40 the odd modes are below 5×10^{-8}, the $l = 2$ tail holds to ρ_{out} and the $l = 4$ tail to $\rho \approx 20$, and M_{ADM}, M_2, M_4 are good to 10^{-6}, 2×10^{-5} and 2×10^{-3} relative. In the mass-centered harmonic chart $c_l\rho^{l+1}$ tends to 0.58387, 0.1747 and 0.0220, i.e. to M_{ADM}, M_2 and M_4, consistently with harmonic coordinates being asymptotically Cartesian and mass centered [31, 32]: a second, independent route to the same numbers.

VII. CONCLUSIONS

A PINN solves the exterior boundary-value problem of the static vacuum Einstein equations with prescribed inner data and asymptotic flatness, and reproduces exact solutions with and without symmetry. Two ingredients are essential: a network whose connection is derived from its metric, and a complete gauge fixing. Harmonic coordinates and the inner data leave the asymptotic chart free to translate and to rotate; these modes are invisible to every residual except at a very low level, and they are the largest error until the pseudo-Newtonian center of mass and its antisymmetric counterpart, the rotation flux, are fixed. With that, the boundary conditions are minimal: λ and the induced metric on the inner sphere, pointwise Robin conditions on the outer one, nothing on h_{rr} or h_{rA}; conditions added beyond these only make the solution worse. We expect the same issue, and the same remedy, in any elliptic formulation of the stationary problem whose inner data do not fix the asymptotic frame. The Geroch–Hansen moments of the computed solutions can be read off the symmetry axis without constructing Weyl coordinates, and they are insensitive to that translation. The extension to the stationary equations is the natural next step.

Appendix A: Reproducing the results

[TODO: repository URL and the commit the paper's numbers come from.] The code is the `Stationary` package of that repository: `stationary/train.py` (the solver; every option below is a flag of `python -m stationary.train`), `stationary/specref.py` (the spectral reference), `stationary/gh_moments.py` (the moments). Software: Python 3.14, JAX 0.11.1, Flax 0.12.9, Optax 0.2.8, in double precision (`JAX_ENABLE_X64=1`); hardware: NVIDIA A30 (24 GB). Each result is produced by the commands below, all in the repository.

- Exact-solution checks: `tests/test_pipeline.py`, `verify_weyl.py`, `tests/test_gh_moments.py`.
- Schwarzschild tests (Table I): `Paper/commands_schwarzschild.sh`.
- One recipe: `Paper/hub_recipe.sh` holds the formulation and training used for every network run of Secs. IV C–VI (it reproduces the configuration of `ab_none`), and its phases produce the Weyl tests (`wr_*`: Table II, Fig. 1, the rotated test `wr_rot45`), the off-axis networks at width 40 or 20 with 16384 or 32768 points (`rp_*`), the larger spectral reference ($l \le 14$) and the regrading against it, the further initializations (`sd1_*`, `sd2_*`) and the hard end at 65536 points (`hs_*`).
- Quadrupolar data: `recipe_pq16k.sh` (the problem). With and without the center of mass (Table III, Fig. 4), `ab_none_*` in `Paper/hub_qp.sh` and `fn_nocom_*` in `Paper/hub_qp5.sh` (Adam 3 minutes, 10–11 minutes per SSBroyden stage on one A30), graded against `runs/ab_none_w2/specref_nonlinear_fine.npz` (`Paper/hub_mass.sh`). The runs with extra conditions on the outer spheres, `ab_*` in

TABLE V. Mass and quadrupole moment of the network (the best one for each amplitude; width in the third column, empty where none was trained) and of the spectral reference, with $M_0 = R_0 = 1/\sqrt{3} = 0.577350$. The spectral references use $l \le 10$ and 36 radial modes for $|S_2| \le 1/6$, $l \le 14$ and 48 radial modes at $-1/4$ and $l \le 20$ and 48 at -0.30. The spread of the mass over the fit windows of Sec. VI is below 10^{-4} in $M_{\rm ADM}$ up to $S_2 = -1/6$, and 1.1×10^{-3} and 4.2×10^{-3} at $-1/4$ and -0.30.

			$M_{\rm ADM}$			M_2	
S_2	$\lambda_{\min}$	width	network	spectral ref.	$(M_{\rm ADM}^{\rm spec} - M_0)/S_2^2$	network	spectral ref.
-0.30	0.033	40	0.767355	0.766484	2.101	1.3558	1.4072
$-1/4$	0.083	40	0.669611	0.669610	1.476	0.8067	0.8067
$-1/6$	0.167	20	0.607969	0.607970	1.102	0.4052	0.4052
$-1/8$	0.208	–	–	0.593055	1.005	–	0.2793
$-1/12$	0.250	20	0.583847	0.583847	0.936	0.1747	0.1747
$-1/24$	0.292	20	0.578886	0.578886	0.885	0.0833	0.0833
$+1/12$	0.417	–	–	0.582915	0.801	–	-0.1540

`Paper/hub_qp.sh`, and `qp_com_*`: its Adam stage in `mendieta_submit.sh` (round P), its SSBroyden stages in `Paper/hub_qp.sh` (20 minutes each with the extra outer spheres). Off-axis data, `Paper/hub_qp3.sh` (`qp_rot3_*`, width 20, 10 minutes per stage) and `Paper/hub_qp4.sh` (`qp_rot4_n32768_*`, width 40, 30 minutes per stage). The mass against S_2 (Table V), `Paper/hub_mass.sh`: the networks `ms_s2h_*` ($S_2 = -1/24$) and `ms_s2d_*` ($-1/6$), and the spectral reference of each amplitude (`specref_nonlinear_fine` in the run directory; `Paper/commands_specref.sh` for its settings, 35–40 minutes each). The hard end: the networks `ms_s2_025_*` and `ms_s2_030_*` continued by `Paper/hub_hard2.sh`, at width 40 with 32768 points `Paper/hub_hard3.sh` (`*_b40_*`) and with 65536 points `Paper/hub_recipe.sh` (`hs_*`, Table V); the spectral references with the damped step of `stationary/specref.py`, `Paper/hub_hardref.sh` ($-1/4$, tag `damp`) and `Paper/hub_bigref.sh` (-0.30, tag `big_l20`; 12–40 minutes per Newton step on one A30, 6 GB of device memory).

- The width-60 off-axis runs and the rotation condition (Sec. V C, Fig. 5): phases I (`rp_w60_n65k_one`), J (`rp_w60_n65k_tan_one`) and L (`rp_w60_n65k_rot`) of `Paper/hub_recipe.sh`, and phase M (`wr_rot45_rot`) for the rotated Weyl continuation; the mode analysis `scripts/error_modes.py`, the flux identities `scripts/green_identities.py`, the figure `Paper/fig_training.py`. The Weyl figure (Fig. 1): `python -m stationary.plane runs/wr_prod_w2 --surface`.
- The geometry figures (Figs. 2 and 3): `scripts/geometry_quadrupole.py` on the run `ms_s2_025_w5` with the spectral reference `specref_nonlinear_damp.npz` (about 5 minutes for the slice and one hour for the Hawking profile on a laptop CPU).
- The spectral reference of Sec. V: `stationary/specref.py` with the settings of `SPEC` in `Paper/hub_mass.sh` (loss 1.6×10^{-13}, 6 minutes on one A30); the larger one, phase C of `Paper/hub_recipe.sh`.
- Grading and moments: `scripts/grade_rotated.py`, `scripts/fresh_sample_loss.py`, `scripts/gh_moments_runs.py`, `scripts/tail_decay.py`.

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